

# CSE402: Numerical Analysis, Simulation and Modeling

## Sessional

Section: C2

Duration: 25 Minutes

---

### When the Generator Fails the Test: Hidden Period Collapse and the K–S Test

#### Background:

Consider the multiplicative congruential generator (MCG):

$$X_{n+1} = (a \cdot X_n) \bmod m, \quad U_n = \frac{X_n}{m}, \quad \text{with } m = 2^{16} = 65536, \ a = 5, \ X_0 = 1$$

Note that  $c = 0$  here, so the Hull–Dobell full-period conditions do not apply. A generator can still look perfectly uniform on paper while being structurally broken underneath.

#### Task 1

Implement the following function. It should return a list of  $U_n$  values.

```
def lcg_generator(n_samples, m=65536, a=5, X0=1):  
    # your implementation
```

#### Task 2

Perform a one-sample, two-sided Kolmogorov–Smirnov (K–S) test to evaluate whether the generated stream adheres to  $H_0 : U_n \sim \text{Uniform}(0, 1)$  vs.  $H_1 : U_n \not\sim \text{Uniform}(0, 1)$ .

#### Steps:

1. **Generate Stream:** Produce  $N = 100,000$  values  $(U_1, U_2, \dots, U_N)$  using `lcg_generator`.
2. **Sort Variates:** Sort the sequence in ascending order such that  $U_{(1)} \leq U_{(2)} \leq \dots \leq U_{(N)}$ .
3. **Compute Empirical CDF:** Compute the two-sided K–S statistic  $D_N$ :

$$D_N = \max_{1 \leq i \leq N} \left\{ \max \left( \frac{i}{N} - U_{(i)}, U_{(i)} - \frac{i-1}{N} \right) \right\}$$

4. **Evaluate Critical Threshold:** Calculate the large-sample critical value at significance level  $\alpha = 0.05$ :

$$D_{\text{crit}} = \frac{1.36}{\sqrt{N}}$$

5. **Decision Rule:** Compare  $D_N$  against  $D_{\text{crit}}$  and explicitly state whether to accept or reject  $H_0$ .

#### Task 3

You should find that the test in Task 2 fails to reject  $H_0$ —the marginal distribution appears to be uniform. Answer the following questions:

- (a) Calculate the period of the MCG programmatically (in code). Does your result match the theoretically expected value for this MCG?
- (b) Given that the period is  $p$  and  $p \ll N$ , explain concretely what happens to the sequence  $U_n$  for  $n > p$ , and why this behavior does not manifest as a violation of uniformity in a one-sample K–S test.